

⚡ 10,000 – 15,000 AZN GRANT & ACCELERATOR DECK

v1.231-RC1 PRODUCTION READY

IronWaves POS Platform

The Next-Generation Multi-Tenant Cloud POS & Hospitality Operating System

Sub-100ms register latency, real-time Kitchen Display (KDS), mobile table ordering, and Apple iOS loyalty ecosystem built for high-throughput modern dining & retail.

1,450+

PRODUCTION COMMITS

< 100ms

TOUCH RESPONSE TIME

3+

LIVE COMMERCIAL VENUES

Founder: **Abbas Aliyev** (@Enjoyer09) · Live Demo: demo.ironwaves.store

⚠️ THE PROBLEM

Hospitality Tech is Fragmented, Slow & Expensive

Merchants are crippled by tool sprawl, hardware lock-in, and lost revenue during peak hours.

🧩 Tool Fragmentation

Restaurants juggle 4–6 disconnected vendors: POS register, kitchen tickets, loyalty club, mobile floor plan, and accounting.

Cost: \$400–\$900/month/location with frequent data sync failures.

🐢 Crippling Checkout Lag

Legacy systems lag 2–4 seconds per item tap or screen transition. Slow checkouts create long queues and customer walk-outs.

Impact: 15–25% lost revenue during peak dining rushes.

🔒 Hardware Lock-In

Legacy vendors force merchants into proprietary \$2,000–\$5,000 terminals with zero cross-platform flexibility.

Impact: Prohibitive capital expenses for multi-branch expansion.

💡 THE SOLUTION

The Unified Cloud-Native Operating System

One integrated platform replacing multiple legacy tools with blazing speed.

- ⚡ **Sub-100ms Touch Register:** Instant cart actions, barcode scanning, multi-tender split payments (Cash, Card, Points, Staff).
- ⚡ **Smart Multi-Station KDS:** Real-time WebSocket ticket routing to Bar, Hot Line, and Cold Prep with auditory rush alerts.
- ⚡ **Waiter Mobile Pad:** 1-hand mobile table ordering, seat check splitting, and course firing right at the dining table.
- ⚡ **Apple iOS V3 Loyalty PWA:** Digital member passes with fraud-proof dynamic QR tokens and Starbucks-style store selection.
- ⚡ **Fiscal Shift Integrity:** Automated X-Reports and end-of-day Z-Reports with precise double-entry cash drawer reconciliation.



Device Agnostic Architecture

Runs seamlessly on any browser, iPad, Android tablet, touchscreen POS terminal, or smartphone without proprietary hardware fees.

Web SPA

iOS Native

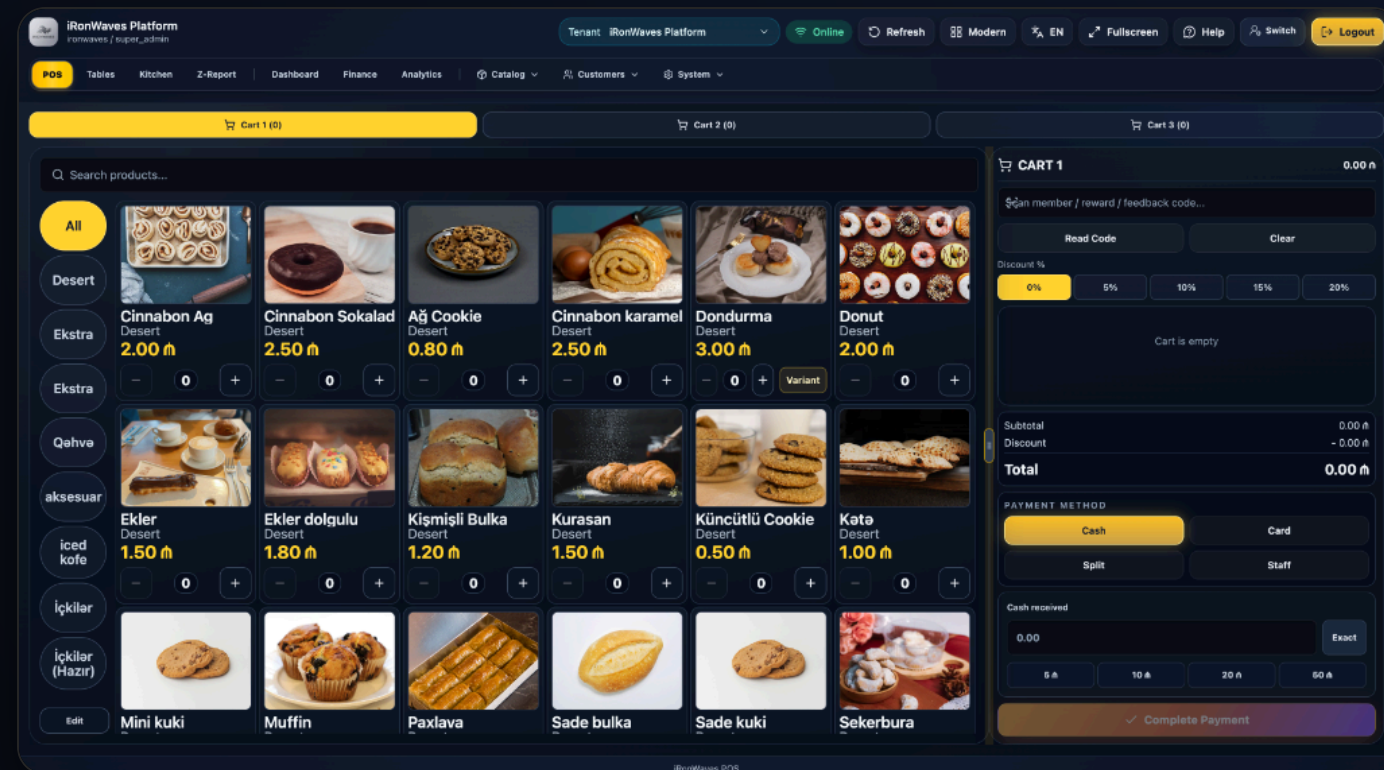
Android Native

Capacitor 8

ESC/POS Thermal

High-Speed Touch POS & Multi-Cart Register

Engineered for intense checkout velocity during morning coffee & lunch queues.



Multi-Cart Tabs (Cart 1, 2, 3)

Cashiers can park an order when a guest hesitates and instantly serve the next customer with zero context loss.

Flexible Split Tenders

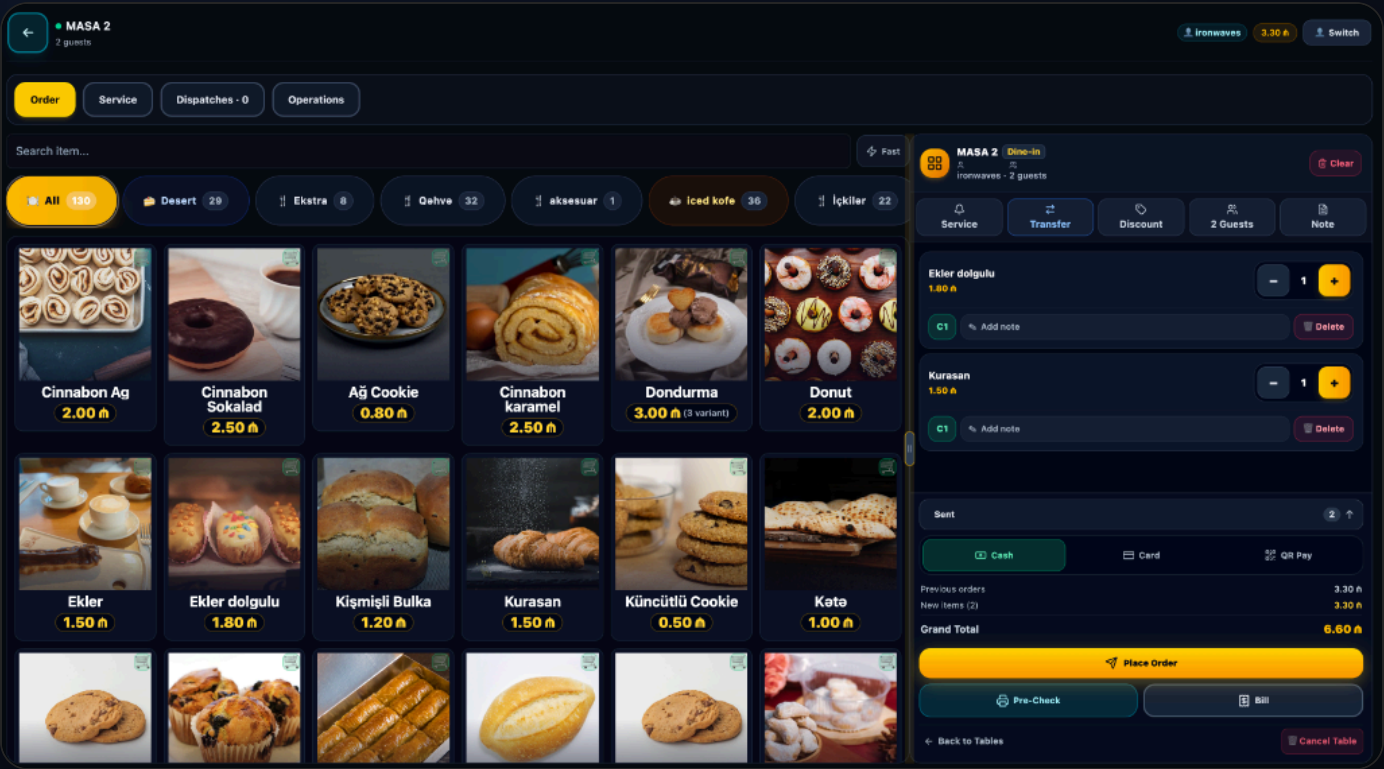
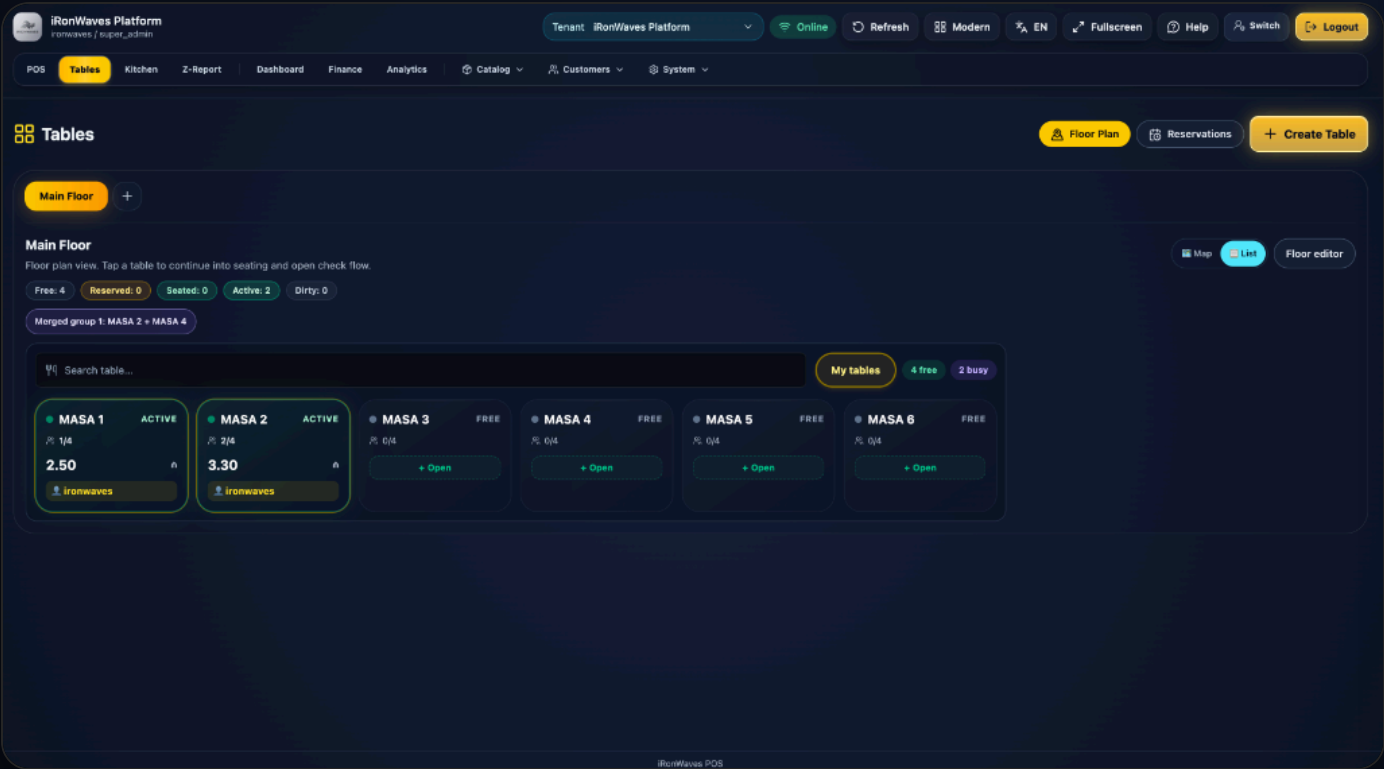
Seamlessly split bills between Cash, Credit Card, Loyalty Points, Gift Cards, and Staff accounts in a single tap.

Direct Silent Thermal Printing

Automated ESC/POS generation via QZ Tray and native background print agent for Windows & macOS.

Interactive Floor Plan & Seat-Level Dine-In

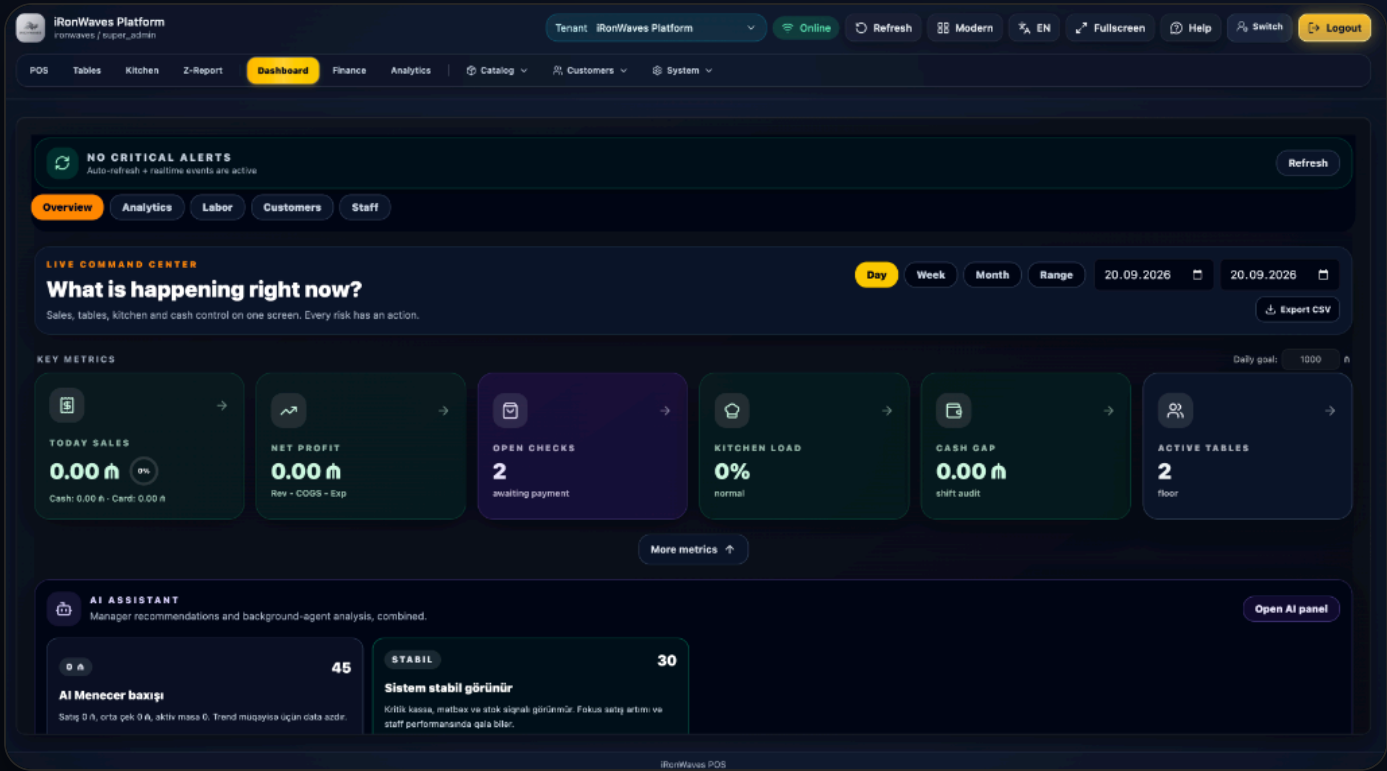
Turn tables 25% faster with mobile order taking and course-by-course kitchen pacing.



Left: Real-time table occupancy, merge groups, and reservations · Right: Waiter pad with seat-specific notes, guest picker & bill splitting.

Live Command Center & AI Manager

Real-time visibility into revenue, kitchen load, cash variance, and smart recommendations.



Real-Time Metric Telemetry

Instant tracking of Today's Sales, Net Profit Margins, Open Checks, Kitchen Load %, and Cash Drawer Variance.



Autonomous AI Manager

Embedded AI Copilot evaluates hourly sales velocity, flags abnormal discount usage, and automates re-ordering alerts.

Swift Terminal Access & Digital Loyalty Pass

Role-based PIN switching and Apple iOS-inspired customer loyalty.



12 34 4-Digit Quick PIN Switching

Rapid staff switching prevents wrong order attribution and tracks individual staff performance metrics.

📱 Apple V3 Customer Loyalty PWA

Cryptographic dynamic QR member pass, tier progression (Bronze → Silver → Gold), and Starbucks-style store selection.

 MARKET OPPORTUNITY

A \$29 Billion Industry Undergoing Cloud Modernization

Independent restaurants and regional chains are rapidly ditching legacy hardware.

\$29.09B

TAM: GLOBAL POS SOFTWARE (2028)

12.8% CAGR driven by cloud migration and contactless ordering.

\$8.4B

SAM: INDEPENDENT F&B & CAFÉS

Specialty coffee, bakeries, casual dining & cloud kitchens.

\$140M

SOM: TARGET REGIONAL ROLLOUT

Fast-growing multi-branch hospitality groups in underserved regions.

 Ideal Initial Beachhead

Specialty coffee shops and boutique bakeries — venues with high daily transaction frequency, urgent queue pressure, and strong customer loyalty requirements.

Predictable SaaS Subscription + Transparent Pricing

All-inclusive single pricing per location with zero hardware lock-in or hidden costs.

ALL-INCLUSIVE PRIMARY VENUE

55 AZN/month

- ⚡ **1 Primary Venue / Location:** Complete platform access
- ⚡ **Ultra-Fast POS Cashier:** Sub-100ms checkout interaction
- ⚡ **Kitchen Display System (KDS):** Unlimited station screens
- ⚡ **Waiter Mobile Table Pads:** Interactive visual floor plan
- ⚡ **Customer Loyalty App:** Apple V3 cryptographic QR pass
- ⚡ **Thermal Print Agent:** Native direct USB/LAN/Bluetooth print
- ⚡ **Real-time Financials:** Shift management & X/Z fiscal reports

MULTI-BRANCH EXPANSION

+39 AZN/month / extra venue

- ⚡ **Each Additional Location:** Scalable multi-branch expansion
- ⚡ **Live Cross-Branch Stock:** Real-time synchronized inventory
- ⚡ **Centralized Chain Analytics:** Unified business dashboard & audits
- ⚡ **Aggregated Financials:** Consolidated franchise P&L reporting
- ⚡ **Cross-Location Staff Roaming:** Secure 4-digit PIN switching
- ⚡ **Zero Hardware Mandate:** Runs on any iPad, Android, or PC
- ⚡ **Dedicated Priority Support:** 24/7 technical hotline & onboarding

📈 High-Margin Expansion Streams

Payment Processing Margin: 0.20%–0.35% transaction fee on integrated local QR & card rails (m10 / Bank APIs) · **Native Mobile Store Publishing:** \$199 setup for branded iOS App Store & Google Play Store publishing · **Turnkey Local AI Appliance:** Private on-premise voice ordering server.

How IronWaves Wins the Market

Faster, hardware-independent, and 60% lower total cost of ownership.

CAPABILITIES	LEGACY (IIKO, MICROS)	STANDARD (SQUARE, CLOVER)	ENTERPRISE (TOAST)	IRONWAVES ⚡
Touch Latency	Slow (2–4s)	Medium (500ms)	Medium (400ms)	Ultra-Fast (<100ms)
Hardware Agnostic	❌ Proprietary	❌ Locked Terminals	❌ Locked Terminals	✅ Any Browser / iPad / Android
Multi-Tenant Cloud	❌ On-Premise	⚠️ Single Account	⚠️ Limited Multi-Org	✅ Native Subdomains (*.store)
Integrated KDS & Tables	⚠️ Extra Fees	⚠️ Basic	✅ Good	✅ Native Live WebSockets
Customer Loyalty PWA	❌ Third-party SMS	⚠️ Basic	⚠️ Add-on Fees	✅ Apple iOS V3 QR Wallet
Monthly Cost / Location	\$250–\$600+	\$80–\$180+	\$165–\$350+	\$49–\$129 (60% Lower TCO)

 TRACTION & VALIDATION

Battle-Tested Engineering in Production Venues

Not a mockup — a fully functional system powering real merchant operations.

Active Commercial Merchants

- ⚡ **gyrospos.ironwaves.store:** High-throughput fast-casual dining venue in Baku.
- ⚡ **emalatxana.ironwaves.store:** Artisan bakery & dine-in restaurant.
- ⚡ **super.ironwaves.store:** Enterprise administrative hub & franchise manager.

Interactive Public Sandbox

Live sandbox at demo.ironwaves.store allows anyone to test cashier orders, tables, and KDS with pre-loaded mock menus and zero signup.

1,450+

CLEAN GIT COMMITS

Continuous, iterative production engineering with 100% solo founder velocity.

v1.231-RC1

OFFICIAL RELEASE CANDIDATE

Full release tagged on GitHub with precompiled print agent binaries for Windows and macOS.

🎯 THE ASK & 6-MONTH MILESTONES

Target Funding: 10,000 – 15,000 AZN Grant

Local LLM & Edge AI R&D, multi-tenant cloud scaling, and native store distribution.

🎯 Strategic Fund Allocation

- ⚡ **45% (5,000 – 7,000 AZN) — Local LLM & Edge AI Workstation:** Dedicated Apple Silicon workstation (Mac Studio Unified Memory) to run, quantize, and fine-tune offline LLMs (Llama 3, Gemma, Whisper) for POS voice ordering, multi-lingual receipts, and inventory AI with <100ms offline latency and zero cloud API fees.
- ⚡ **25% (2,500 – 3,500 AZN) — Multi-Tenant Cloud & Database:** High-availability PostgreSQL, Redis pub/sub live ordering sync, automated backups, and wildcard subdomains on Railway.
- ⚡ **15% (1,500 – 2,500 AZN) — Native Store Distribution:** Apple Developer Program & Google Play Store publishing, Capacitor 8 iOS/Android native packaging, and automated CI/CD for waiter pads.
- ⚡ **15% (1,000 – 2,000 AZN) — Local Payments & Merchant Testbed:** Direct API integrations with local payment rails (m10 QR acquiring, Kapital Bank / PASHA Bank) and pilot printer kits.

📅 6-Month Execution Milestones

- ⚡ **Month 1–2 (Local AI & Cloud):** Mac Studio R&D testbed setup, offline quantized model testing, Railway DB high-availability hardening.
- ⚡ **Month 3–4 (Native Stores & Payments):** App Store & Google Play Store release, m10 QR payments, onboarding 15+ specialty cafés.
- ⚡ **Month 5–6 (Scale & Edge Pilots):** Scale to 25+ paying venues, deploy offline AI voice ordering in pilot restaurant, cash flow positive.

Let's Build the Future of Commerce ⚡

Abbas Aliyev · Founder & Lead Engineer

demo.ironwaves.store · github.com/Enjoyer09/ironwaves-pos-platform